Jonas El Gammal

Quantitative researcher | Applied ML & data systems | Scientific software developer

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PROFILE

Quantitative researcher and scientific software developer working across probabilistic machine learning, computational physics, and healthcare software. I turn complex models and fragmented data into tested systems for inference, automation, and decision-making. My work spans product strategy, system architecture, numerical implementation, validation, and delivery.

SELECTED TECHNICAL IMPACT

- Led the technical design of an integrated platform that turns fragmented practice data into automated workflows, patient-facing services, and operational insight.
- Delivered simulation workflows, code-quality assurance, and computational-statistics methods for LISA, a joint ESA & NASA space mission.
- Built and maintained open-source Python/JAX packages for surrogate inference, differentiable gravitational-wave modelling, astrophysical population simulation, and sampler interoperability.
- Built tool-using AI agents and agent loops with structured plans, deterministic validation, traceable execution, and bounded repair.

EXPERIENCE

Co-founder, Managing Director & CTO - Bright Interface GmbH 2026-present
- Lead product and technical strategy, architecture, engineering quality, data integration, applied AI, and delivery.

Postdoctoral Researcher - KBFI / NICPB, Tallinn 2026-present
- Develop reusable statistical methods and production-quality research software for extracting reliable conclusions from complex, noisy data.

Postdoctoral Researcher - University of Insubria 2025-2026
- Built GPU-accelerated, differentiable inference software for large time-series datasets within an international scientific collaboration.

Visiting Researcher - CERN 2024
- Built and validated statistical software for reconstructing weak signals in simulated data for LISA.

SELECTED PROJECTS

GPry	Probabilistic ML	Gaussian-process surrogate modelling, failure classification, active learning, convergence diagnostics, and parallel acquisition for expensive Bayesian inference.
SIGWAY	Differentiable modelling	Modular JAX framework connecting primordial physics, cosmological kernels, numerical integration, and Bayesian reconstruction.
fastropop	Stochastic simulation	Reproducible simulation of black-hole-binary populations, spectra, and skymaps for population-level statistical studies.
Graphsmith	Reliable AI systems	Experimental typed graph planning with deterministic compilation, validation, local repair, and traceable execution.

METHODS AND TECHNOLOGY

Methods: Bayesian inference, Gaussian processes, active learning, uncertainty quantification, Monte Carlo methods, numerical optimisation, stochastic simulation, time-series and signal analysis, MCMC, nested sampling, differentiable programming, model validation.

AI systems: LLM agents, tool use, agent loops, multi-agent workflows, planning and execution, structured outputs, prompt and context design, validation, repair, tracing, and evaluation harnesses.

Engineering: Python, JAX, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, and C++; FastAPI and REST APIs; Svelte; Docker; OAuth/OIDC; encrypted data workflows; Git, Linux, CI/CD, HPC/GPU computing, testing, documentation, and releases.

EDUCATION

PhD in Physics - University of Stavanger 2021-2025
Developed Gaussian-process-accelerated Bayesian inference and primordial-spectrum reconstruction methods for LISA; taught full courses in Numerical Methods and Electromagnetism.

MSc and BSc in Physics - RWTH Aachen University 2015-2020
Astroparticle physics and cosmology; MSc thesis on Gaussian-process acceleration of expensive Bayesian inference.

LANGUAGES

German - native | **English** - near-native | **French** - B2/C1